Survival of *Phellinus weirii* in Douglas-fir stumps after logging[1]

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Stumps and roots of trees harvested 20, 30, and 50 years ago were examined for survival of *Phellinus weirii*. Survival was compared among sites regenerated to *Pseudotsuga menziesii* (Douglas-fir) and *Alnus rubra* (red alder). Distinctive patterns of decay and the persistence of setal hyphae and zone lines after death of the fungus enabled documentation of survival time and of the retreat of *P. weirii* from roots. *Phellinus weirii* often died from a decay column without colonizing adjacent sound wood.

Data from 1076 roots on 340 stumps were analyzed. After 50 years, the fungus survived in 26% of the stumps and in 11% of the roots sampled. Surface mycelium was present on 17% of the roots after 20 years, 15% after 30 years, and 1% after 50 years. The decrease in survival time was similar on sites presently occupied by red alder or by Douglas-fir. Survival was not correlated with stump diameter or degree of deterioration after 20 and 30 years. In the current stand, trees killed by *P. weirii* were found within 5 m of 45% of infected stumps after 50 years.

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Les souches et les racines d'arbres coupés 20, 30 et 50 ans ont été examinées pour déterminer le degré de survivance de *Phellinus weirii* sur des sites occupés par une régénération de *Pseudotsuga menziesii* (sapin Douglas) ou d'*Alnus rubra* (aulne rouge). L'aspect caractéristique de la carie et la persistance, même après la mort du champignon, de setae et d'une ligne noire bordant la carie ont servi d'indices pour établir la longueur de la période de survie, ou le moment de la disparition de *P. weirii* des racines. *Phellinus weirii* était souvent disparu de la colonne de carie sans avoir du coloniser le bois sain adjacent.

Les données provenant de 1076 racines et de 340 souches furent analysées. Après 50 ans, le champignon était encore vivant dans 26% des souches et 11% des racines examinées. Du mycélium de surface était présent sur 17, 15 et 1% des racines où le champignon avait survécu respectivement 20, 30 et 50 ans. Il n'y avait pas de différence dans la diminution du taux de survie sur les sites occupés par l'aulne rouge ou le sapin Douglas. La survivance du champignon après 20 et 30 ans n'était pas reliée au diamètre de la souche ni à son degré de décomposition. Dans le peuplement actuel, plusieurs arbres tués par *P. weirii* ont été trouvés à moins de 5 m de 45% des souches encore infectées après 50 ans.

[Traduit par le journal]

Introduction

Phellinus weirii (Murr.) Gilb. causes laminated root rot of Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco) and other conifers in western North America. The fungus spreads slowly by mycelial growth across root contacts and survives between tree generations in the roots and stumps of dead infected trees (Hansen 1976; Wallis and Reynolds 1965). Attempts to inoculate intact and wounded trees and stumps with basidiospores have consistently failed (Nelson 1971) and spores are not thought to play an important role in disease spread (Childs 1970).

Recommended control for laminated root rot is exhaustion or disruption of the saprophytic survival capability of the fungus by crop rotation with a resistant species or by mechanical stump removal (Hadfield and Johnson 1977). Red alder (*Alnus rubra* Bong.) is the resistant species most often recommended because of its nitrogen fixing capabilities, rapid growth, and commercial value. Also, alder and its symbionts produce phenolic and fatty acid compounds inhibitory to *P. weirii* in vitro (Li et al. 1969), which has led to an hypothesis that alder exerts an active biological control of root rot (Trappe 1972; Nelson et al. 1978).

Practicality of crop rotation or mechanical stump removal depends largely on how long the fungus

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TABLE 1. Characteristics of stands in which *P. weirii* infected conifer stumps were excavated

Location	Years since harvest	Current overstory	No. of stumps sampled	Average stump diameter, cm
Sweet Home, OR	19–21	Douglas-fir	21	80
Blue River, OR	20	Douglas-fir	23	85
Alsea, OR	20	Douglas-fir	13	72
Mapleton, OR	20	Douglas-fir	10	78
Sweet Home, OR	21–22	Alder	14	60
Sweet Home, OR	23–24	Alder	23	60
Mapleton, OR	22	Alder	21	82
Alsea, OR	20	Alder	23	84
Grand Ronde, OR	30–31	Douglas-fir	23	109
Lowell, OR	31	Douglas-fir	16	78
Mapleton, OR	30	Douglas-fir	8	133
Blue River, OR	28	Douglas-fir	18	106
McCleary, WA	50	Douglas-fir	19	133
Vernonia, OR	43–50	Douglas-fir	14	139
Dexter, OR	52	Douglas-fir	25	100
Corvallis, OR	47–57	Douglas-fir	24	94
McCleary, WA	50	Alder	24	100
Vernonia, OR	43–50	Alder	21	115

NOTE: OR, Oregon; WA, Washington.

survives in old stumps and roots. *Phellinus weirii* has been found alive in stumps 50 years and more after logging, but quantitative data on survival have been limited to shorter times (Wallis and Reynolds 1965). This study initially measured fungus survival in 20-year-old stumps (Hansen 1976), and has been expanded to include 30- and 50-year-old stumps and to compare survival on sites occupied, since logging, by red alder and Douglas-fir. The 20-year data are retabulated here for comparison with 30- and 50-year data.

Materials and Methods

Douglas-fir and western hemlock stumps from harvests approximately 20, 30, and 50 years ago were dug in 12 stands of Douglas-fir and red alder in western Oregon and Washington, U.S.A. (Table 1). Selection was based on the date of previous logging, evidence of root rot in the previous stand, and distribution in the Cascade and Coast Ranges. Study sites were located in cooperation with the United States Department of Agriculture Forest Service and private timber companies.[3]

The characteristic white pocket rot caused by *P. weirii* was distinguished from decay caused by other root or butt rot fungi in coastal Douglas-fir by (1) the concentration of disease in infection centers marked by wind-thrown root-rotted trees or their remains; (2) hollow stumps or roots; (3) crescent-shaped cavities continuing from stumps into roots; (4) black zone lines; (5) wood separating along annual rings (lamination); and (6) setal hyphae. Thick-walled, reddish brown setal hyphae, visible with a hand lens, were indicative of *P. weirii* and often persisted after the fungus had died. Other single characteristics were not proof of past *P. weirii* infection but in combination were probably reliable indicators.

Stumps to be analyzed were located in two stages on sites occupied, after logging, by red alder or Douglas-fir. First, a stump with obvious signs of past infection was located and the surrounding stumps examined for aboveground signs of *P. weirii*. Three roots on each stump suspected of infection were excavated with hand tools for their entire length or to a depth of about 50 cm. Stumps and excavated roots were then carefully examined for evidence of past *P. weirii* activity. Only those with positive indications were used in data tabulations.

Roots were examined for pattern of decay, evidence of past *P. weirii* infection, presence of living *P. weirii*, and ectotrophic mycelium of *P. weirii*. The diameter and length of each root at the margin between living and dead *P. weirii* were recorded. At least one piece of each root was removed for isolation on malt agar; more were taken if apparently viable mycelium with setal hyphae was found. *Phellinus weirii* was identified in culture by its characteristic setal hyphae.

Results

Sapwood was often nearly gone, but heartwood and bark of most stumps appeared unaltered 20 and 30 years after harvest. Even after 50 years, massive amounts of sound wood remained in the center of many old stumps. Roots of 50-year-old stumps were more deteriorated, although bark was often still intact.

Stumps 20, 30, and 50 years after harvest showed progressive retreat of *P. weirii* from infected roots toward the stump. Zone lines frequently surrounded

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TABLE 2. Survival of *Phellinus weirii* in conifer stumps and roots 20, 30, and 50 years after logging on red alder and Douglas-fir sites

Years since harvest	Current overstory	No. of stumps sampled	Live *P. weirii* in stumps, %	No. of roots sampled	Live *P. weirii* in roots, %	Roots with surface mycelium, %	Root diameter,[a] cm (no.)[b]	Distance[a] from stump, m (no.)[b]
20	Douglas-fir	67	96	211	73	17	13 (68)	1.1 (71)
20	Alder	81	75	278	63	25	13 (111)	1.6 (109)
30	Douglas-fir	65	59	216	39	15	19 (54)	1.5 (60)
50	Douglas-fir	82	26	237	11	1	33 (16)	0.9 (18)
50	Alder	45	56	134	19	2	36 (14)	0.6 (20)

[a]At furthest margin between living and dead *P. weirii*.
[b]Number of stumps sampled is in parentheses.

live *P. weirii* and were still evident after the death of the fungus. Series of old zone lines marked the retreat of the fungus from the distal portion of roots. On some roots with collapsed bark, zone lines formed the only barrier between an island of living *P. weirii* and the soil. Most often, however, the fungus survived only in roots with intact bark.

Little evidence appeared of the advance of *P. weirii* into sound wood after tree harvest. Often the fungus died from a decay column and left a crescent-shaped hollow surrounded by sound wood. A resin-soaked band often separated the old decay column from uninfected wood. In several 30-year-old stumps *P. weirii* survived only in the bark of partially hollow roots after the substrate in the old decay column had been exhausted. After 50 years it was found alive in the bark only once, but old, crescent-shaped *P. weirii* decay columns bounded by uninfected wood were regularly encountered. Often one decay column was found interior to another, with no sign of decay in the intervening solid wood. Decay columns with dead or living *P. weirii* occupied roughly the same proportion of roots and stumps after 50 years as after 20 years. The proportion of stumps more than 50% hollow did not increase significantly with stump age; at 20, 30, and 50 years after harvest it was 16%, 14%, and 17%, respectively.

Phellinus weirii was often difficult to isolate from wood in which it appeared viable, especially in older stumps where the fungus survived in thin columns between resin-soaked bands or in small islands within zone lines. In these cases removing an uncontaminated sample for isolation was difficult. Cultural confirmation of visual estimates of viability decreased with stump age. *Phellinus weirii* was isolated from 87% of the stumps in which it was judged viable after 20 years, from 78% after 30 years, and from 63% after 50 years. Only once was *P. weirii* isolated from a sample judged not viable. Table 2 is based on visual determinations of viability in 340 stumps. No consistent differences in survival showed between stands in different areas, so data are summarized by stump age and current overstory.

On sites regenerated to Douglas-fir, *P. weirii* was alive in 96%, 59%, and 26% of the stumps after 20, 30, and 50 years, respectively. Survival in individual roots decreased even more strikingly (Table 2). On sites regenerated to red alder, survival decreased similarly, although only 20- and 50-year-old stumps were examined. The margin between living and dead *P. weirii* occurred in increasingly large roots with increasing stump age. After 20 years, the fungus was recovered 13 times from roots less than 1 cm in diameter. At 50 years, the smallest root in which *P. weirii* was alive was 10 cm in diameter. The fungus survived in an island enveloped by a zone line. Live *P. weirii* in roots extended an average of less than 2 m from stumps after 20 years and less than 1 m after 50 years (Table 2). Decayed roots sometimes ended in callus tissue near the stumps. On one site, 23% of the infected roots were broken and callused, and overall, 7% of the roots were shortened in this way. Portions of roots distal to old breaks were occasionally recovered, and always showed advanced deterioration and signs of dead *P. weirii*.

Ectotrophic mycelium was infrequently present on roots of any age and then only in isolated patches. It was found on only 5 of 371 roots of 50-year-old stumps (Table 2). Roots of several 30-year-old stumps had surface mycelium only where they contacted a rock or the root of a living tree.

No significant difference in survival appeared between sites occupied by red alder or Douglas-fir, although alder sites showed more variation. After 20 years, there was less *P. weirii* overall on alder sites, but the two sites with highest survival in stumps, individual roots, and surface mycelium were in red alder stands. After 50 years, *P. weirii* survival was greater in stumps in alder stands than in Douglas-fir

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TABLE 3. Survival (%) of *Phellinus weirii* in stumps with different degrees of decay and diameters

Years since harvest	Current overstory	Stumps		Stump diameter class		
		10% hollow	50% hollow	<0.5 m	<1 m	>1 m
20	Douglas-fir	87 (38)[a]	100 (12)	100 (8)	94 (50)	100 (16)
20	Alder	77 (60)	75 (12)	85 (20)	77 (70)	70 (10)
30	Douglas-fir	56 (45)	67 (9)	43 (7)	52 (23)	63 (40)
50	Douglas-fir	30 (37)	36 (14)	0 (1)	25 (32)	26 (47)
50	Alder	70 (30)	13 (8)	— (0)	40 (20)	69 (26)

[a]Number of stumps sampled is in parentheses.

stands, but since stumps were excavated in only two alder stands of that age (Table 1) the difference must be interpreted cautiously. Survival in individual roots, presence of surface mycelium, and diameter of the root at the furthest margin between living and dead *P. weirii* were nearly identical in 50-year-old alder and fir stands.

Survival of *P. weirii* after 30 years did not correlate with stump diameter, but only one stump less than 0.5 m in diameter was excavated on the 50-year sites. There was no difference in survival in stumps hollowed by *P. weirii* compared with stumps with less advanced decay through 30 years, but in 50-year red alder stands, survival was reduced in hollow stumps (Table 3).

Phellinus weirii was active in the current generation in all Douglas-fir stands examined but had killed susceptible conifers within a 5-m radius of less than half the stumps (Table 4). More mortality appeared around 50-year-old stumps than around younger ones and more around stumps with live *P. weirii* than around stumps in which the fungus had died. Sixty-five percent of 50-year-old stumps had dead current-generation trees within 10 m but proximity of the old infected stumps to each other resulted in considerable double counting.

TABLE. 4. Current generation *Phellinus weirii* caused tree mortality within 5 m of old stumps with living or dead *P. weirii*[a]

Years since harvest	Current overstory	Stumps with associated tree mortality, %	
		P. weirii alive	*P. weirii* dead
20	Douglas-fir	33 (12)[b]	67 (2)
30	Douglas-fir	43 (16)	23 (6)
50	Douglas-fir	57 (12)	41 (22)

[a]Only stumps with living or dead current generation trees within 5 m were tallied.
[b]Number of stumps is in parentheses.

Discussion

This research provides the first quantitative measure of the limits of *P. weirii* survival in old stumps. Results should be useful for planning and evaluating control strategies, but limitations must be understood.

Roots growing deeper than about 50 cm were not regularly sampled because of the method of excavation. The margin between viable and dead *P. weirii* was not uncovered on many roots. Nelson (1967) has shown that *P. weirii* survival in buried cubes of decayed wood increases with depth; *P. weirii* may survive longer in deep roots also.

Fungus survival varied considerably between stands which is not surprising considering the varied environments and stump sizes sampled. Stands harvested 50 years ago were generally at lower elevations in easily accessible areas. More recent logging that left the 20- and 30-year-old stumps used in this study was at higher elevations, often on steep terrain. No consistent differences were evident between the northern and southern stands or between those in the Cascade and Coast Ranges.

Precise logging history was not available for two of the 50-year-old stands. One had greater and the other less survival of *P. weirii* than the 50-year-old stands with known history. Limited survival of *P. weirii* was confirmed, however, by examination of 40- to 50-year-old infected stumps pushed from the ground with bulldozers in two additional stands (E. M. Hansen, unpublished data; W. Thies, United States Department of Agriculture Forest Service, Pacific Northwest Forest and Range Experiment Station, personal communication). In most of these stumps the fungus was no longer viable and in the remainder it was confined to the largest roots and the lower part of the stump.

Strict criteria for selection of stumps for analysis and the reliance on visual rather than cultural determination of viability were necessary for a rigorous

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study. Any bias they may introduce is toward survival of the fungus. *Phellinus weirii* may once have been present in stumps rejected from the sample because past infection could not be confirmed. This is particularly true of western hemlock (*Tsuga heterophylla* (Raf.) Sarg.) stumps. Old *Fomes annosus* (Fr.) Cooke infections could not be distinguished from old decay caused by *P. weirii* unless setal hyphae remained. Twenty-one western hemlock stumps in the sample had positive indications of *P. weirii*, but many others were rejected for lack of certainty. Among those included, the degree and pattern of *P. weirii* survival did not differ from survival in Douglas-fir stumps.

Finally, this study measured fungus survival, but success of a crop rotation scheme will depend on fungus infectivity when a susceptible tree is replanted. *Phellinus weirii* spreads as ectotrophic mycelium across root contacts. Since it disappears more rapidly from root surfaces than from within roots (Table 2), old stumps may be less infectious than survival figures suggest. This is supported by the apparent precarious condition of the fungus within old stumps, often just a small interior island within zone lines. Though evidence of encroachment of other fungi on *P. weirii* was frequently found, *P. weirii* was not found advancing over wood already decomposed by other organisms. Limited infectivity is further shown by the lack of mortality around many old stumps. On the other hand, observations that the ectotrophic mycelium survives only at contact points may mean that it can be reestablished by host or contact stimulus. This must be explored further. The carry-over of infections from one stand generation to the next is currently being studied.

This study has implications for the management of infected conifer stands in the Pacific Northwest. The threat of new infections from old stumps seems to be nearly gone after 50 years. Because roots of few young trees extend to infected residual roots in the 10 years after planting, and because small roots of those young trees that are killed decompose rapidly, the period of effective infectivity may be shorter still. The length of rotation with a resistant crop will be determined by balancing economic considerations against the decreasing risk, with time, of renewed *P. weirii* infection.

Old infection centers often fill with shrubs that sprout after fire and logging. Rehabilitation of these areas by scarification is expensive if massive old-growth stumps must be removed to reduce root rot inoculum. If logging is 40 years or more past, however, removal should not be necessary.

Though red alder is a good resistant species in a crop rotation scheme, apparently it does not shorten the time required to reduce the inoculum. By far the most vigorous and extensive development of ectotrophic mycelium was on infected roots of 20-year-old stumps in alder stands established immediately after logging on the Sweet Home District of the Willamette National Forest. Alder root nodules were found adjacent to every old Douglas-fir root excavated in these stands.

The apparent lack of correlation between size and degree of deterioration of stumps and survival of *P. weirii* (Table 3) is puzzling, although a trend may be developing in the 50-year-old stumps. Small stumps may be lacking in the 50-year-old stands because they have completely decomposed or small trees may not have been harvested. Attempts to assess infection hazard from individual stumps will require further careful evaluation.

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